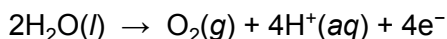
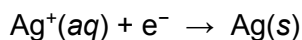


The electrolysis of aqueous silver nitrate involves two separate half-reactions that occur at the anode (Reaction 1) and the cathode (Reaction 2).



Reaction 1



Reaction 2

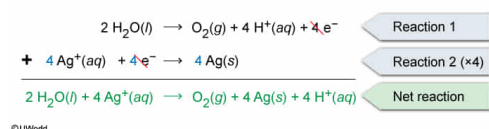
Which of the following net reactions occurs during the electrolysis?

- ☐ A. $2\text{H}_2\text{O}(l) + \text{Ag}^+(aq) \rightarrow \text{O}_2(g) + \text{Ag}(s) + 4\text{H}^+(aq)$ [14%]
- ☐ B. $2\text{H}_2\text{O}(l) + 4\text{Ag}^+(aq) \rightarrow \text{O}_2(g) + 4\text{Ag}(s) + 2\text{H}_2(g)$ [7%]
- ☐ C. $2\text{H}_2\text{O}(l) + \text{Ag}(s) \rightarrow \text{O}_2(g) + \text{Ag}^+(aq) + 4\text{H}^+(aq)$ [4%]
- ✓ ☒ D. $2\text{H}_2\text{O}(l) + 4\text{Ag}^+(aq) \rightarrow \text{O}_2(g) + 4\text{Ag}(s) + 4\text{H}^+(aq)$ [72%]

Correct

73% answered correctly

Explanation:



For an oxidation-reduction (redox) reaction to be balanced, the same number of each type of atom and the same net charge must be present on both sides of the reaction equation. **Redox reactions** can be expressed as two **half-reactions** (one for the **oxidation** and one for the **reduction**, which occur together). Adding the two half-reactions together returns the **net reaction**. Species present on both sides of the summation cancel out. In a balanced scenario, the total number of electrons released from one half-reaction will match the total number of electrons used by the second half-reaction, causing the electron terms to cancel completely. If not, **multiples** of one (or both) of the half-reactions may be used in the summation.

The net reaction resulting from Reactions 1 and 2 can be found by adding these half-reactions together. In order for the electron terms to fully cancel out, a multiple of Reaction 2 (all terms multiplied by 4) must be used. Canceling the 4e^- terms and summing the reactants side and the products side of the half-reactions yield a net reaction with both an **atom balance** and a **net charge balance** (+4 on each side).

(Choice A) The net charge is not balanced. The net charge is +1 on the reactants side of the reaction, but it is +4 on the products side.

(Choice B) H_2 gas is not a valid product in the net reaction because it is not found in either half-reaction. The net reaction must result from adding multiples of the half-reactions. The net charge is also not balanced.

(Choice C) The $\text{Ag}^+(aq)$ and $\text{Ag}(s)$ terms are on the wrong sides of the equation. This answer results if the reverse of Reaction 2 is added with Reaction 1, but the electron terms **do not cancel** using this approach.

Educational objective:

Redox reactions are fully balanced when the same number of each type of atom and the same net charge are present on both sides of the reaction equation. For redox processes expressed as two half-reactions, the net reaction is found by adding the two half-reactions (or multiples of them) together so that the electron terms cancel.

Time spent:0 sec**Subject:**
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Solutions and Electrochemistry